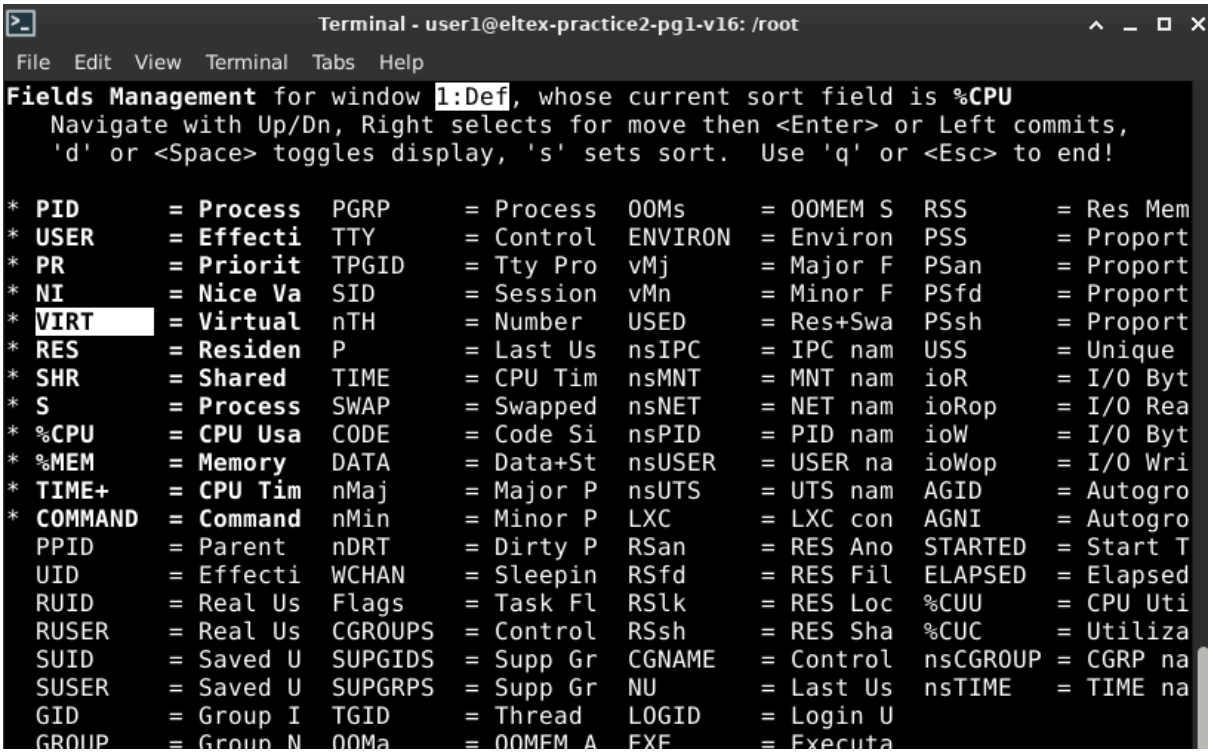
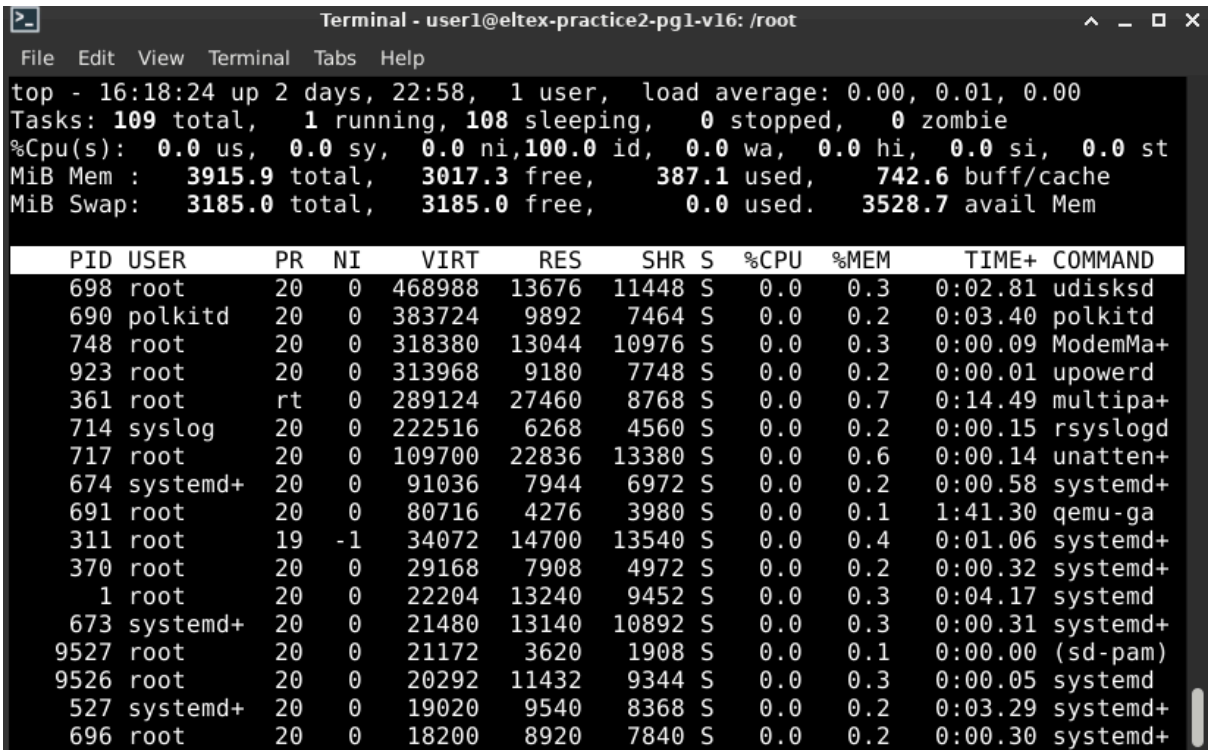
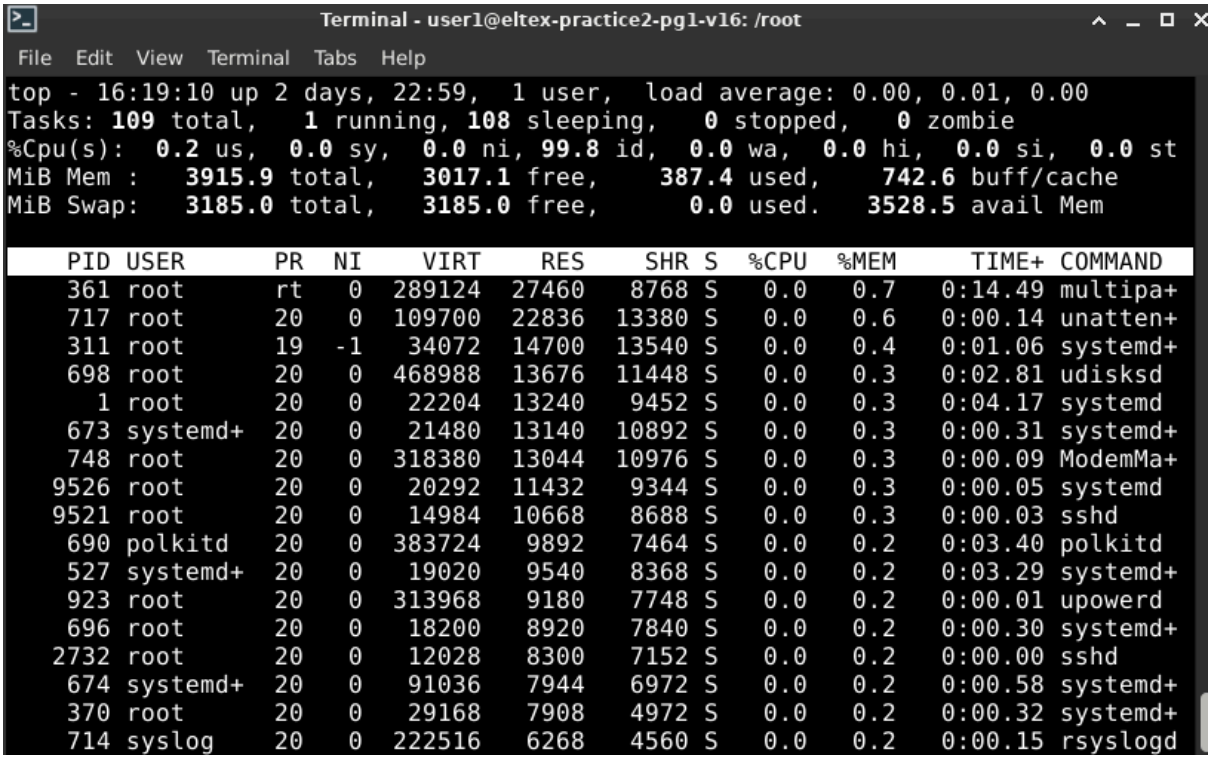
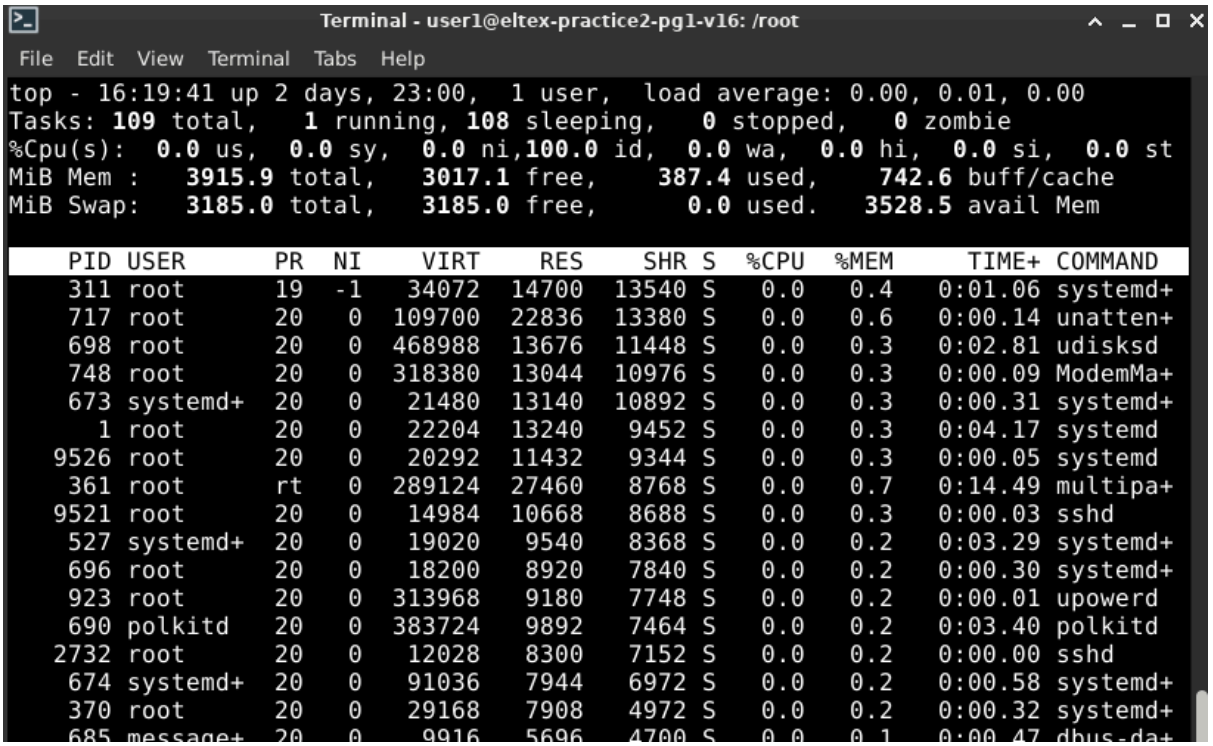
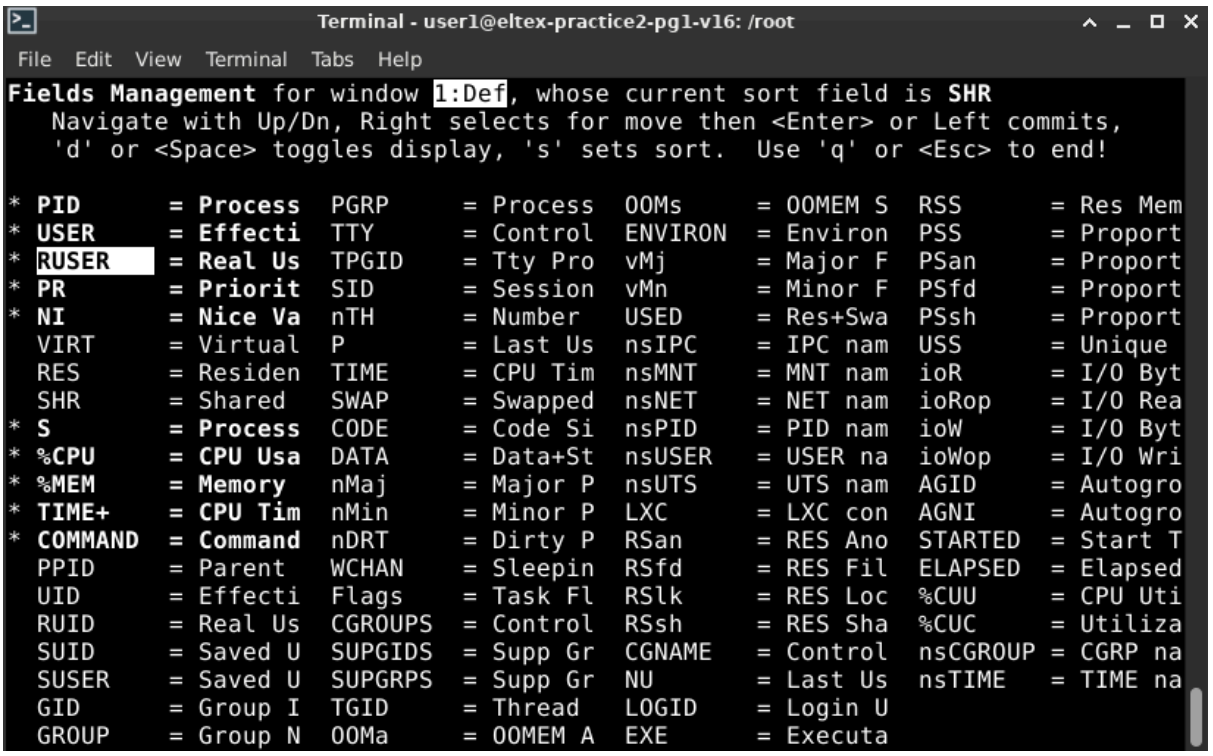

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
root@eltex-practice2-pg1-v16:~# su user1
Warning: your password will expire in 3 days.
user1@eltex-practice2-pg1-v16:/root$ ps -eo nlwp | awk '$1 > 1 {count++} END {pr
int count}
> '
11
user1@eltex-practice2-pg1-v16:/root$

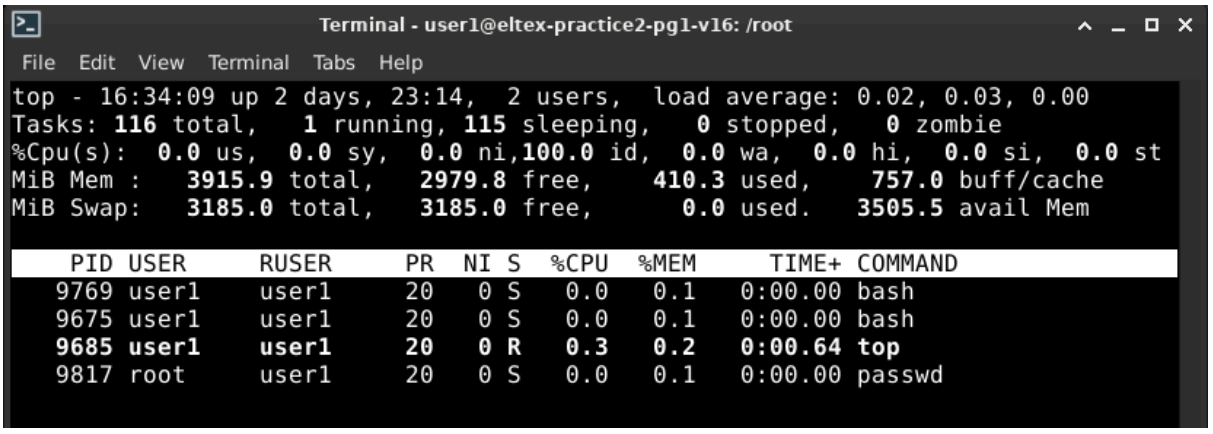

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
Fields Management for window 1:Def, whose current sort field is %CPU
Navigate with Up/Dn, Right selects for move then <Enter> or Left commits,
'd' or <Space> toggles display, 's' sets sort. Use 'q' or <Esc> to end!
* PID = Process PGRP = Process OOMs = OOMEM S RSS = Res Mem
* USER = Effecti TTY = Control ENVIRON = Environ PSS = Proport
* PR = Priorit TPGID = Tty Pro vMj = Major F PSan = Proport
* NI = Nice Va SID = Session vMn = Minor F PSfd = Proport
* VIRT = Virtual nTH = Number USED = Res+Swa PSsh = Proport
* RES = Residen P = Last Us nsIPC = IPC nam USS = Unique
* SHR = Shared TIME = CPU Tim nsMNT = MNT nam ioR = I/O Byt
* S = Process SWAP = Swapped nsNET = NET nam ioRop = I/O Rea
* %CPU = CPU Usa CODE = Code Si nsPID = PID nam ioW = I/O Byt
* %MEM = Memory DATA = Data+St nsUSER = USER na ioWop = I/O Wri
* TIME+ = CPU Tim nMaj = Major P nsUTS = UTS nam AGID = Autogro
* COMMAND = Command nMin = Minor P LXC = LXC con AGNI = Autogro
PPID = Parent nDRT = Dirty P RSan = RES Ano STARTED = Start T
UID = Effecti WCHAN = Sleepin RSfd = RES Fil ELAPSED = Elapsed
RUID = Real Us Flags = Task Fl RSlk = RES Loc %CUU = CPU Uti
RUSER = Real Us CGROUPS = Control RSsh = RES Sha %CUC = Utiliza
SUID = Saved U SUPGIDS = Supp Gr CGNAME = Control nsCGROUP = CGRP na
SUSER = Saved U SUPGRPS = Supp Gr NU = Last Us nsTIME = TIME na
GID = Group I TGID = Thread LOGID = Login U
GROUP = Group N OOMa = OOMEM A EXE = Executa

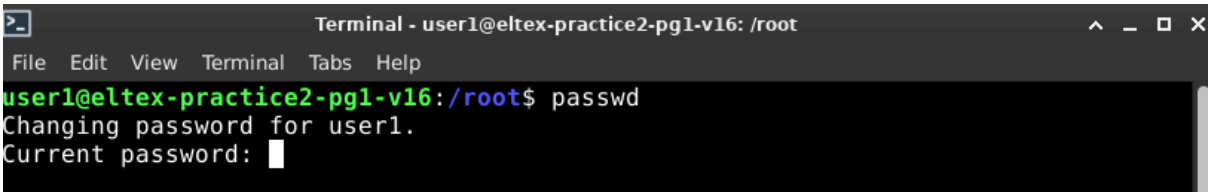

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
top - 16:18:24 up 2 days, 22:58, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 109 total, 1 running, 108 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 3017.3 free, 387.1 used, 742.6 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3528.7 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
698 root 20 0 468988 13676 11448 S 0.0 0.3 0:02.81 udisksd
690 polkitd 20 0 383724 9892 7464 S 0.0 0.2 0:03.40 polkitd
748 root 20 0 318380 13044 10976 S 0.0 0.3 0:00.09 ModemMa+
923 root 20 0 313968 9180 7748 S 0.0 0.2 0:00.01 upowerd
361 root rt 0 289124 27460 8768 S 0.0 0.7 0:14.49 multipa+
714 syslog 20 0 222516 6268 4560 S 0.0 0.2 0:00.15 rsyslogd
717 root 20 0 109700 22836 13380 S 0.0 0.6 0:00.14 unatten+
674 systemd+ 20 0 91036 7944 6972 S 0.0 0.2 0:00.58 systemd+
691 root 20 0 80716 4276 3980 S 0.0 0.1 1:41.30 qemu-ga
311 root 19 -1 34072 14700 13540 S 0.0 0.4 0:01.06 systemd+
370 root 20 0 29168 7908 4972 S 0.0 0.2 0:00.32 systemd+
1 root 20 0 22204 13240 9452 S 0.0 0.3 0:04.17 systemd
673 systemd+ 20 0 21480 13140 10892 S 0.0 0.3 0:00.31 systemd+
9527 root 20 0 21172 3620 1908 S 0.0 0.1 0:00.00 (sd-pam)
9526 root 20 0 20292 11432 9344 S 0.0 0.3 0:00.05 systemd
527 systemd+ 20 0 19020 9540 8368 S 0.0 0.2 0:03.29 systemd+
696 root 20 0 18200 8920 7840 S 0.0 0.2 0:00.30 systemd+

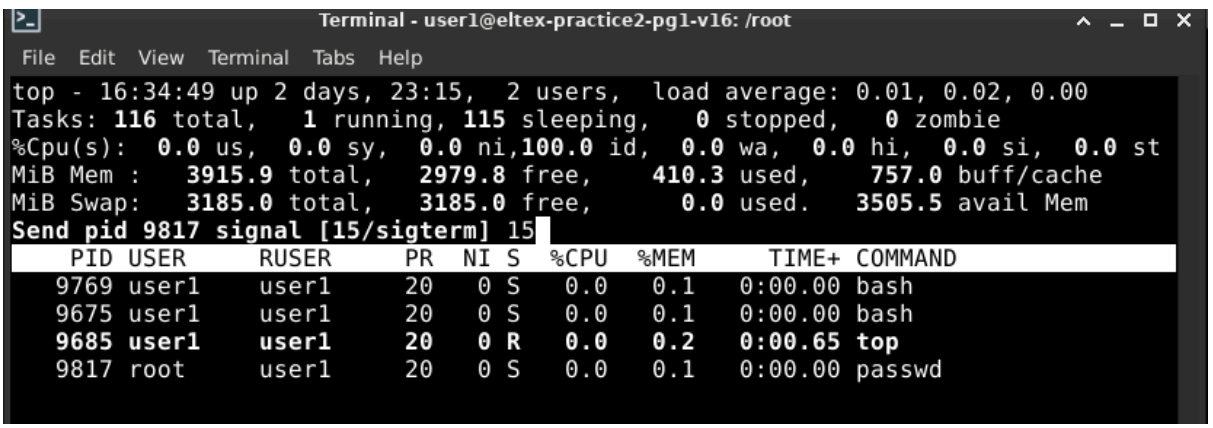

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
top - 16:19:10 up 2 days, 22:59, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 109 total, 1 running, 108 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.0 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 3017.1 free, 387.4 used, 742.6 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3528.5 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
361 root rt 0 289124 27460 8768 S 0.0 0.7 0:14.49 multipa+
717 root 20 0 109700 22836 13380 S 0.0 0.6 0:00.14 unatten+
311 root 19 -1 34072 14700 13540 S 0.0 0.4 0:01.06 systemd+
698 root 20 0 468988 13676 11448 S 0.0 0.3 0:02.81 udisksd
1 root 20 0 22204 13240 9452 S 0.0 0.3 0:04.17 systemd
673 systemd+ 20 0 21480 13140 10892 S 0.0 0.3 0:00.31 systemd+
748 root 20 0 318380 13044 10976 S 0.0 0.3 0:00.09 ModemMa+
9526 root 20 0 20292 11432 9344 S 0.0 0.3 0:00.05 systemd
9521 root 20 0 14984 10668 8688 S 0.0 0.3 0:00.03 sshd
690 polkitd 20 0 383724 9892 7464 S 0.0 0.2 0:03.40 polkitd
527 systemd+ 20 0 19020 9540 8368 S 0.0 0.2 0:03.29 systemd+
923 root 20 0 313968 9180 7748 S 0.0 0.2 0:00.01 upowerd
696 root 20 0 18200 8920 7840 S 0.0 0.2 0:00.30 systemd+
2732 root 20 0 12028 8300 7152 S 0.0 0.2 0:00.00 sshd
674 systemd+ 20 0 91036 7944 6972 S 0.0 0.2 0:00.58 systemd+
370 root 20 0 29168 7908 4972 S 0.0 0.2 0:00.32 systemd+
714 syslog 20 0 222516 6268 4560 S 0.0 0.2 0:00.15 rsyslogd

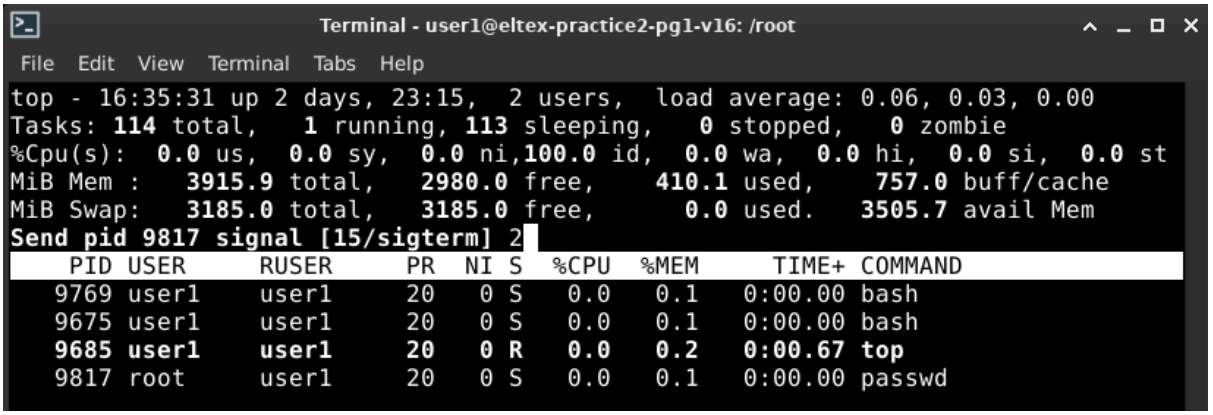

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
top - 16:19:41 up 2 days, 23:00, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 109 total, 1 running, 108 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 3017.1 free, 387.4 used, 742.6 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3528.5 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
311 root 19 -1 34072 14700 13540 S 0.0 0.4 0:01.06 systemd+
717 root 20 0 109700 22836 13380 S 0.0 0.6 0:00.14 unatten+
698 root 20 0 468988 13676 11448 S 0.0 0.3 0:02.81 udisksd
748 root 20 0 318380 13044 10976 S 0.0 0.3 0:00.09 ModemMa+
673 systemd+ 20 0 21480 13140 10892 S 0.0 0.3 0:00.31 systemd+
1 root 20 0 22204 13240 9452 S 0.0 0.3 0:04.17 systemd
9526 root 20 0 20292 11432 9344 S 0.0 0.3 0:00.05 systemd
361 root rt 0 289124 27460 8768 S 0.0 0.7 0:14.49 multipa+
9521 root 20 0 14984 10668 8688 S 0.0 0.3 0:00.03 sshd
527 systemd+ 20 0 19020 9540 8368 S 0.0 0.2 0:03.29 systemd+
696 root 20 0 18200 8920 7840 S 0.0 0.2 0:00.30 systemd+
923 root 20 0 313968 9180 7748 S 0.0 0.2 0:00.01 upowerd
690 polkitd 20 0 383724 9892 7464 S 0.0 0.2 0:03.40 polkitd
2732 root 20 0 12028 8300 7152 S 0.0 0.2 0:00.00 sshd
674 systemd+ 20 0 91036 7944 6972 S 0.0 0.2 0:00.58 systemd+
370 root 20 0 29168 7908 4972 S 0.0 0.2 0:00.32 systemd+

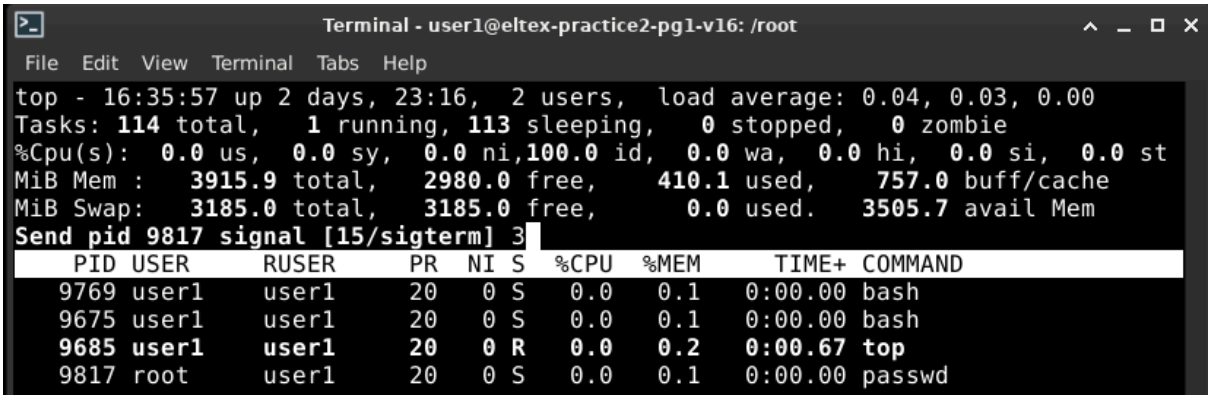

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
Fields Management for window 1:Def, whose current sort field is SHR
Navigate with Up/Dn, Right selects for move then <Enter> or Left commits,
'd' or <Space> toggles display, 's' sets sort. Use 'q' or <Esc> to end!
* PID = Process PGRP = Process OOMs = OOMEM S RSS = Res Mem
* USER = Effecti TTY = Control ENVIRON = Environ PSS = Proport
* RUSER = Real Us TPGID = Tty Pro vMj = Major F PSan = Proport
* PR = Priorit SID = Session vMn = Minor F PSfd = Proport
* NI = Nice Va nTH = Number USED = Res+Swa PSsh = Proport
VIRT = Virtual P = Last Us nsIPC = IPC nam USS = Unique
RES = Residen TIME = CPU Tim nsMNT = MNT nam ioR = I/O Byt
SHR = Shared SWAP = Swapped nsNET = NET nam ioRop = I/O Rea
* S = Process CODE = Code Si nsPID = PID nam ioW = I/O Byt
* %CPU = CPU Usa DATA = Data+St nsUSER = USER na ioWop = I/O Wri
* %MEM = Memory nMaj = Major P nsUTS = UTS nam AGID = Autogro
* TIME+ = CPU Tim nMin = Minor P LXC = LXC con AGNI = Autogro
* COMMAND = Command nDRT = Dirty P RSan = RES Ano STARTED = Start T
PPID = Parent WCHAN = Sleepin RSfd = RES Fil ELAPSED = Elapsed
UID = Effecti Flags = Task Fl RSlk = RES Loc %CUU = CPU Uti
RUID = Real Us CGROUPS = Control RSsh = RES Sha %CUC = Utiliza
SUID = Saved U SUPGIDS = Supp Gr CGNAME = Control nsCGROUP = CGRP na
SUSER = Saved U SUPGRPS = Supp Gr NU = Last Us nsTIME = TIME na
GID = Group I TGID = Thread LOGID = Login U
GROUP = Group N OOMa = OOMEM A EXE = Executa

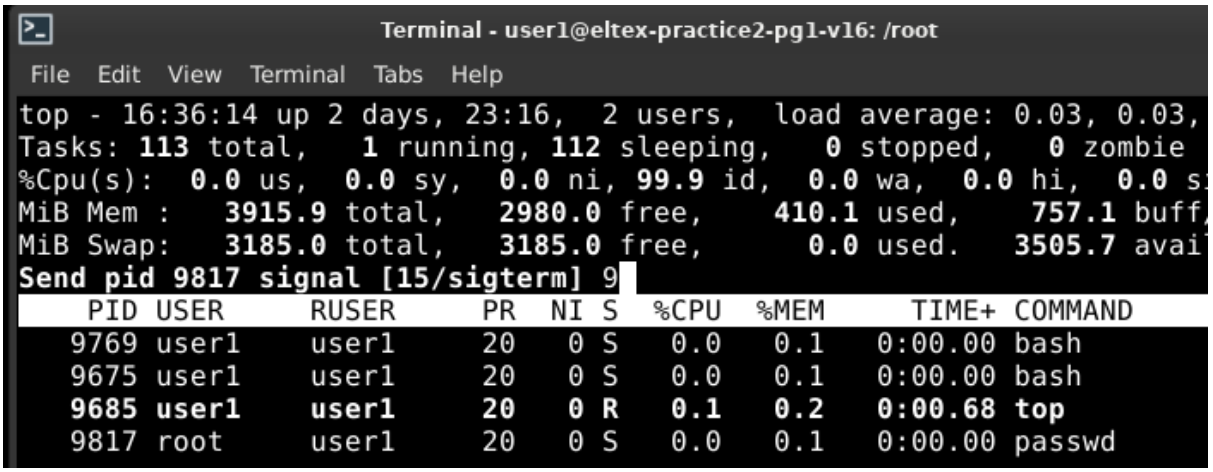

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
top - 16:34:09 up 2 days, 23:14, 2 users, load average: 0.02, 0.03, 0.00
Tasks: 116 total, 1 running, 115 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 2979.8 free, 410.3 used, 757.0 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3505.5 avail Mem
PID USER RUSER PR NI S %CPU %MEM TIME+ COMMAND
9769 user1 user1 20 0 S 0.0 0.1 0:00.00 bash
9675 user1 user1 20 0 S 0.0 0.1 0:00.00 bash
9685 user1 user1 20 0 R 0.3 0.2 0:00.64 top
9817 root user1 20 0 S 0.0 0.1 0:00.00 passwd

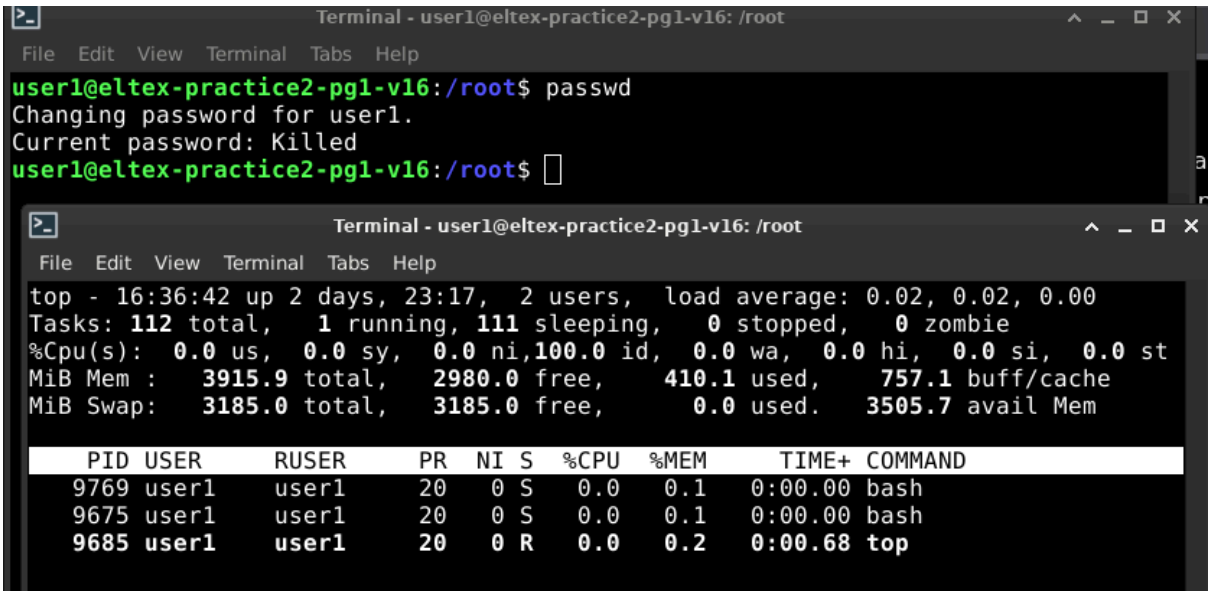

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
user1@eltex-practice2-pg1-v16:/root$ passwd
Changing password for user1.
Current password:

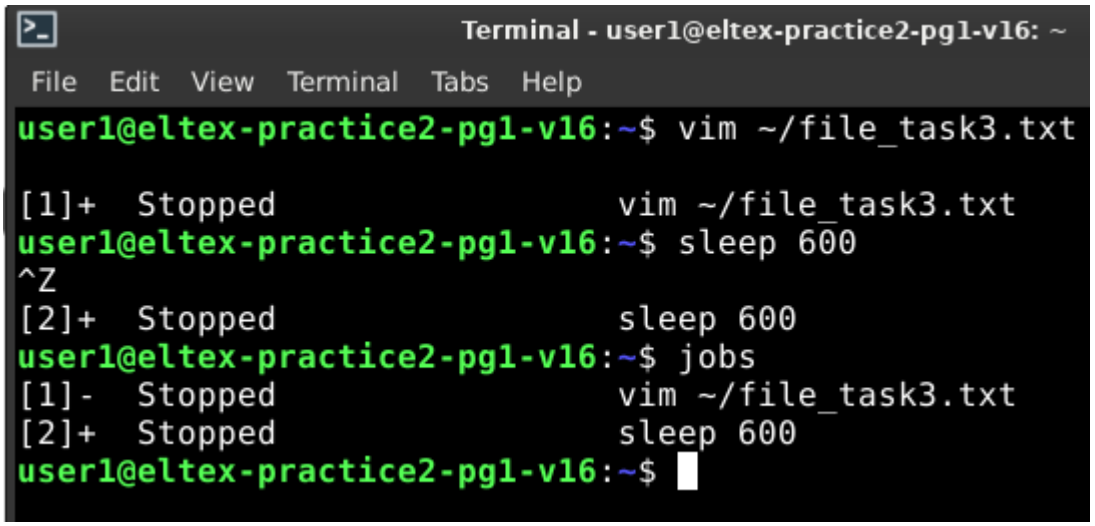

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
top - 16:34:49 up 2 days, 23:15, 2 users, load average: 0.01, 0.02, 0.00
Tasks: 116 total, 1 running, 115 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 2979.8 free, 410.3 used, 757.0 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3505.5 avail Mem
Send pid 9817 signal [15/sigterm] 15
PID USER RUSER PR NI S %CPU %MEM TIME+ COMMAND
9769 user1 user1 20 0 S 0.0 0.1 0:00.00 bash
9675 user1 user1 20 0 S 0.0 0.1 0:00.00 bash
9685 user1 user1 20 0 R 0.0 0.2 0:00.65 top
9817 root user1 20 0 S 0.0 0.1 0:00.00 passwd

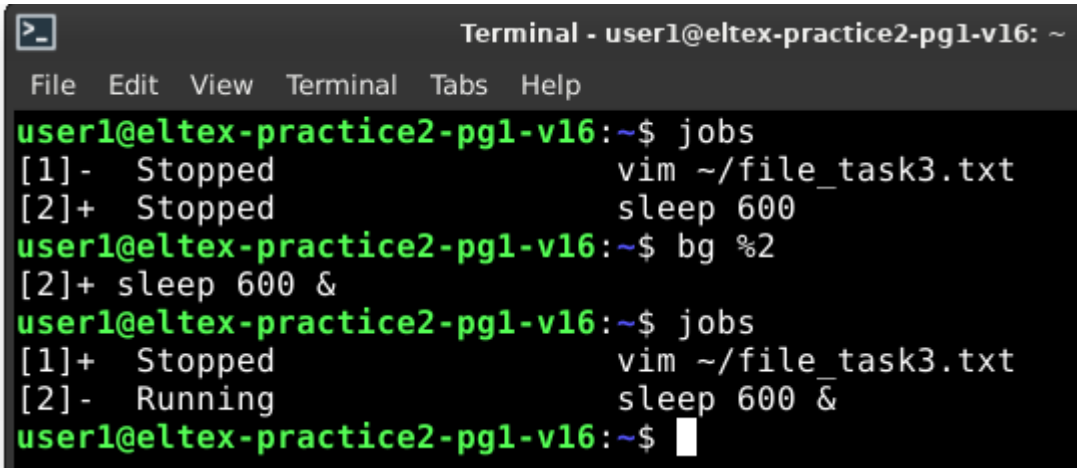

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
top - 16:35:31 up 2 days, 23:15, 2 users, load average: 0.06, 0.03, 0.00
Tasks: 114 total, 1 running, 113 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 2980.0 free, 410.1 used, 757.0 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3505.7 avail Mem
Send pid 9817 signal [15/sigterm] 2
PID USER RUSER PR NI S %CPU %MEM TIME+ COMMAND
9769 user1 user1 20 0 S 0.0 0.1 0:00.00 bash
9675 user1 user1 20 0 S 0.0 0.1 0:00.00 bash
9685 user1 user1 20 0 R 0.0 0.2 0:00.67 top
9817 root user1 20 0 S 0.0 0.1 0:00.00 passwd

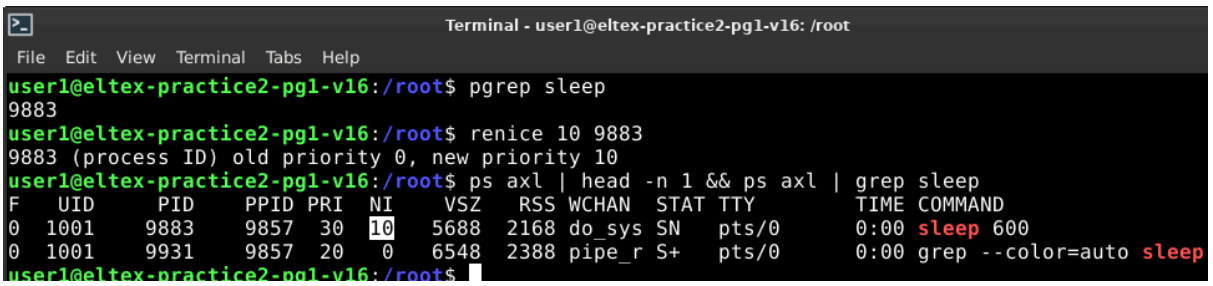

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
top - 16:35:57 up 2 days, 23:16, 2 users, load average: 0.04, 0.03, 0.00
Tasks: 114 total, 1 running, 113 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 2980.0 free, 410.1 used, 757.0 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3505.7 avail Mem
Send pid 9817 signal [15/sigterm] 3
PID USER RUSER PR NI S %CPU %MEM TIME+ COMMAND
9769 user1 user1 20 0 S 0.0 0.1 0:00.00 bash
9675 user1 user1 20 0 S 0.0 0.1 0:00.00 bash
9685 user1 user1 20 0 R 0.0 0.2 0:00.67 top
9817 root user1 20 0 S 0.0 0.1 0:00.00 passwd

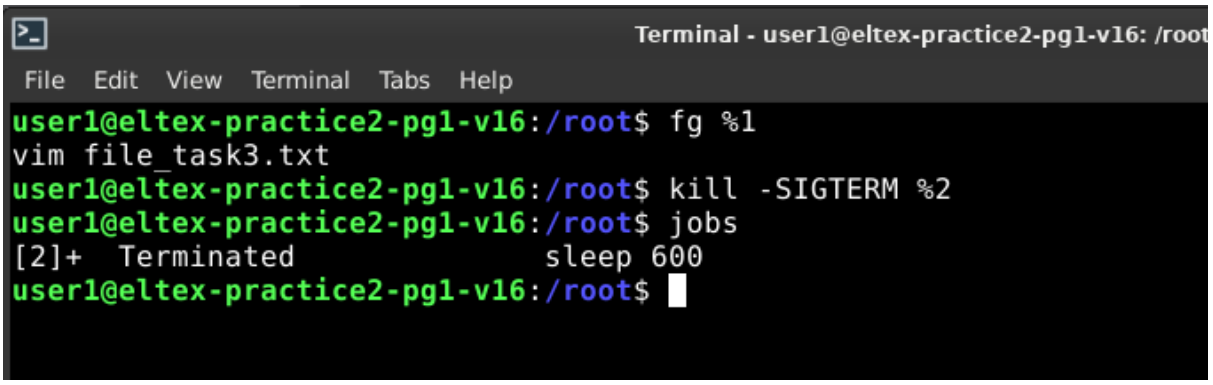

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
top - 16:36:14 up 2 days, 23:16, 2 users, load average: 0.03, 0.03,
Tasks: 113 total, 1 running, 112 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si
MiB Mem : 3915.9 total, 2980.0 free, 410.1 used, 757.1 buff/
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3505.7 avai
Send pid 9817 signal [15/sigterm] 9
PID USER RUSER PR NI S %CPU %MEM TIME+ COMMAND
9769 user1 user1 20 0 S 0.0 0.1 0:00.00 bash
9675 user1 user1 20 0 S 0.0 0.1 0:00.00 bash
9685 user1 user1 20 0 R 0.1 0.2 0:00.68 top
9817 root user1 20 0 S 0.0 0.1 0:00.00 passwd

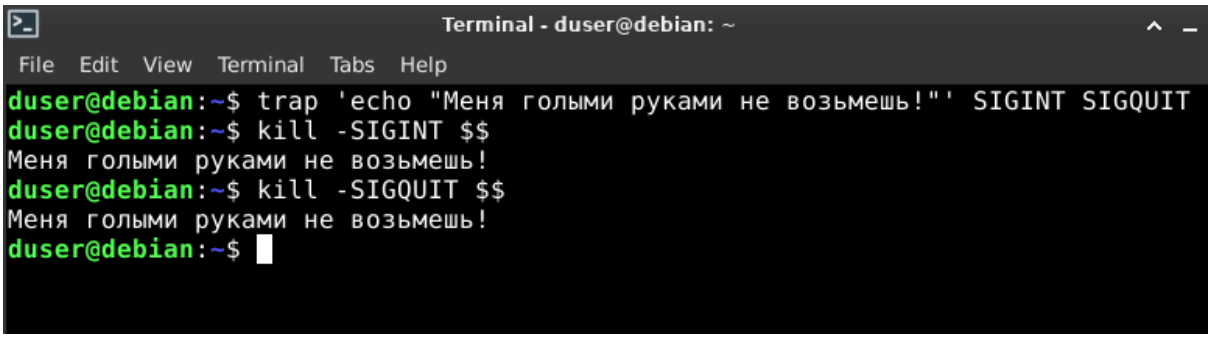

Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
user1@eltex-practice2-pg1-v16:/root$ passwd
Changing password for user1.
Current password: Killed
user1@eltex-practice2-pg1-v16:/root$
Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
top - 16:36:42 up 2 days, 23:17, 2 users, load average: 0.02, 0.02, 0.00
Tasks: 112 total, 1 running, 111 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 2980.0 free, 410.1 used, 757.1 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3505.7 avail Mem
PID USER RUSER PR NI S %CPU %MEM TIME+ COMMAND
9769 user1 user1 20 0 S 0.0 0.1 0:00.00 bash
9675 user1 user1 20 0 S 0.0 0.1 0:00.00 bash
9685 user1 user1 20 0 R 0.0 0.2 0:00.68 top

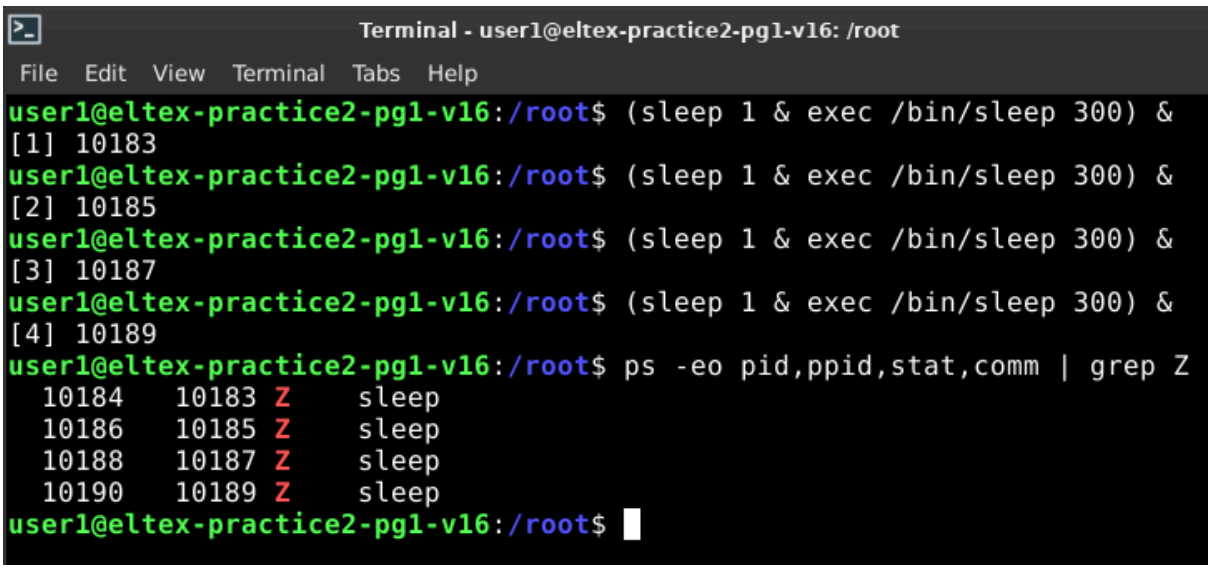

Terminal - user1@eltex-practice2-pg1-v16: ~
File Edit View Terminal Tabs Help
user1@eltex-practice2-pg1-v16:~$ vim ~/file_task3.txt
[1]+ Stopped vim ~/file_task3.txt
user1@eltex-practice2-pg1-v16:~$ sleep 600
^Z
[2]+ Stopped sleep 600
user1@eltex-practice2-pg1-v16:~$ jobs
[1]- Stopped vim ~/file_task3.txt
[2]+ Stopped sleep 600
user1@eltex-practice2-pg1-v16:~$

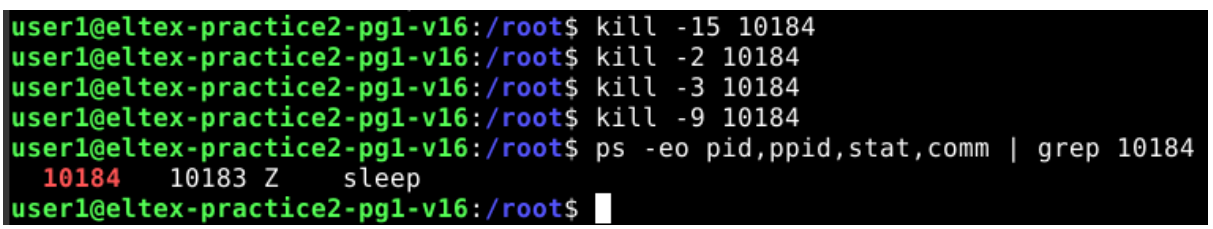

Terminal - user1@eltex-practice2-pg1-v16: ~
File Edit View Terminal Tabs Help
user1@eltex-practice2-pg1-v16:~$ jobs
[1]- Stopped vim ~/file_task3.txt
[2]+ Stopped sleep 600
user1@eltex-practice2-pg1-v16:~$ bg %2
[2]+ sleep 600 &
user1@eltex-practice2-pg1-v16:~$ jobs
[1]+ Stopped vim ~/file_task3.txt
[2]- Running sleep 600 &
user1@eltex-practice2-pg1-v16:~$


Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
user1@eltex-practice2-pg1-v16:/root$ pgrep sleep
9883
user1@eltex-practice2-pg1-v16:/root$ renice 10 9883
9883 (process ID) old priority 0, new priority 10
user1@eltex-practice2-pg1-v16:/root$ ps axl | head -n 1 && ps axl | grep sleep
F UID PID PPID PRI NI VSZ RSS WCHAN STAT TTY TIME COMMAND
0 1001 9883 9857 30 10 5688 2168 do_sys SN pts/0 0:00 sleep 600
0 1001 9931 9857 20 0 6548 2388 pipe_r S+ pts/0 0:00 grep --color=auto sleep


Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
user1@eltex-practice2-pg1-v16:/root$ fg %1
vim file_task3.txt
user1@eltex-practice2-pg1-v16:/root$ kill -SIGTERM %2
user1@eltex-practice2-pg1-v16:/root$ jobs
[2]+ Terminated sleep 600
user1@eltex-practice2-pg1-v16:/root$


Terminal - duser@debian: ~
File Edit View Terminal Tabs Help
duser@debian:~$ trap 'echo "Меня голыми руками не возьмешь!"' SIGINT SIGQUIT
duser@debian:~$ kill -SIGINT $$
Меня голыми руками не возьмешь!
duser@debian:~$ kill -SIGQUIT $$
Меня голыми руками не возьмешь!
duser@debian:~$


Terminal - user1@eltex-practice2-pg1-v16: /root
File Edit View Terminal Tabs Help
user1@eltex-practice2-pg1-v16:/root$ (sleep 1 & exec /bin/sleep 300) &
[1] 10183
user1@eltex-practice2-pg1-v16:/root$ (sleep 1 & exec /bin/sleep 300) &
[2] 10185
user1@eltex-practice2-pg1-v16:/root$ (sleep 1 & exec /bin/sleep 300) &
[3] 10187
user1@eltex-practice2-pg1-v16:/root$ (sleep 1 & exec /bin/sleep 300) &
[4] 10189
user1@eltex-practice2-pg1-v16:/root$ ps -eo pid,ppid,stat,comm | grep Z
10184 10183 Z sleep
10186 10185 Z sleep
10188 10187 Z sleep
10190 10189 Z sleep
user1@eltex-practice2-pg1-v16:/root$


user1@eltex-practice2-pg1-v16:/root$ kill -15 10184
user1@eltex-practice2-pg1-v16:/root$ kill -2 10184
user1@eltex-practice2-pg1-v16:/root$ kill -3 10184
user1@eltex-practice2-pg1-v16:/root$ kill -9 10184
user1@eltex-practice2-pg1-v16:/root$ ps -eo pid,ppid,stat,comm | grep 10184
10184 10183 Z sleep
user1@eltex-practice2-pg1-v16:/root$

```
user1@eltex-practice2-pg1-v16:/root$ kill -15 10183
user1@eltex-practice2-pg1-v16:/root$ kill -2 10183
bash: kill: (10183) - No such process
[1]   Terminated              ( sleep 1 & exec /bin/sleep 300 )
user1@eltex-practice2-pg1-v16:/root$ ps -eo pid,ppid,stat,comm | grep 10184
```

```
user1@eltex-practice2-pg1-v16:/root$ kill -2 10187
user1@eltex-practice2-pg1-v16:/root$ ps -eo pid,ppid,stat,comm | grep 10188
[3]-  Interrupt               ( sleep 1 & exec /bin/sleep 300 )
user1@eltex-practice2-pg1-v16:/root$ ps -eo pid,ppid,stat,comm | grep 10187
```

```
user1@eltex-practice2-pg1-v16:/root$ ps -eo pid,ppid,stat,comm | grep Z
  10190   10189 Z    sleep
user1@eltex-practice2-pg1-v16:/root$ kill -3 10189
user1@eltex-practice2-pg1-v16:/root$ ps -eo pid,ppid,stat,comm | grep 10190
[4]+  Quit                    (core dumped) ( sleep 1 & exec /bin/sleep 300 )
user1@eltex-practice2-pg1-v16:/root$ ps -eo pid,ppid,stat,comm | grep 10189
user1@eltex-practice2-pg1-v16:/root$
```

```
user1@eltex-practice2-pg1-v16:/root$ ps -eo pid,ppid,stat,comm | grep Z
  10230   10229 Z    sleep
user1@eltex-practice2-pg1-v16:/root$ kill -9 10229
user1@eltex-practice2-pg1-v16:/root$ ps -eo pid,ppid,stat,comm | grep 10230
[1]+  Killed                  ( sleep 1 & exec /bin/sleep 300 )
user1@eltex-practice2-pg1-v16:/root$ ps -eo pid,ppid,stat,comm | grep 10229
user1@eltex-practice2-pg1-v16:/root$
```

Зомби процессы – это процессы, которые уже завершились, но запись о них в файловое системе процессорной еще хранится, т.к. не родительский процесс не получил код его завершения. При попытке завершить зомби процесс любым из kill сигналов ничего не выходит, что логично – процесс уже завершен. Утилита ps распознает зомби процессы и помечает их буквой Z. При этом если завершить родительский процесс любым из сигналов – зомби процесс исчезает. На самом деле он переподчиняется другому процессу (скорее всего init/systemd/etc с pid 1), тот процесс получает код завершения зомби и благополучно удаляется из таблицы процессов. Таким образом, для избегания накопления зомби процессов нужно писать корректную их обработку в родителях.